

Machine - Cave

1. Creds

SSH :

- cave_man:cave123man
- tribe_leader:rahasiainimah (ini yang deket root)

2. Foothold

LFI

- set di hardening buat replace semua ../

```
package handlers
import (
    "net/http"
    "os"
    "strings"
    "github.com/zams-putra/my-boot2root/cave/cave/server/internal/middlewares"
    "github.com/zams-putra/my-boot2root/cave/cave/server/internal/models"
)
var logDir = "logs/"
func SetLogDir(dir string) {
    logDir = dir
}
func AdminLogs(w http.ResponseWriter, r *http.Request) {
    claims, ok := r.Context().Value(middlewares.ClaimsKey).(*models.Claims)
    if !ok || !claims.IsAdmin {
        http.Error(w, "Forbidden lau sape mpruy", http.StatusForbidden)
        return
    }
    filename := r.URL.Query().Get("file")
    if filename == "" {
        filename = "app.log"
    }
    clean := strings.ReplaceAll(filename, "../", "")
    fullpath := logDir + clean
    content, err := os.ReadFile(fullpath)
    if err != nil {
        http.Error(w, "File not found", http.StatusNotFound)
        return
    }
}
```

```

    }
    w.Header().Set("Content-Type", "text/plain")
    w.Write(content)
}

```

- especially here

```

clean := strings.ReplaceAll(filename, "../", "")
fullpath := logDir + clean

```

- so kalo bypass with

```

....//....//....//....//....//etc/passwd

```

- doi masih tersisa ../

```

....//, di replace menjadi -> ../
- soalnya bagian ini di ilangin
..[../]/ -> nah itu di cut kan menjadi ../

```

- untuk case ini hardening with

```

clean := filepath.Clean(filename)

```

- untuk LFI nya sendiri itu vuln disini

```

fullpath := logDir + clean

```

- and for hardening, should be like this

```

fullpath := filepath.Join(logDir, clean)

```

XSS Steal Cookie

- di file roasting.jsx untuk page roasting emang di set vuln

```

import { useState, useEffect } from 'react'
import { Link } from 'react-router-dom'
import { api } from '../lib/api'
import { useAuth } from '../lib/auth'

```

```

import styles from './Roasting.module.css'

function timeAgo(dateStr) {
  const diff = (Date.now() - new Date(dateStr)) / 1000
  if (diff < 60) return 'just now'
  if (diff < 3600) return `${Math.floor(diff / 60)}m ago`
  if (diff < 86400) return `${Math.floor(diff / 3600)}h ago`
  return `${Math.floor(diff / 86400)}d ago`
}

function initials(name) {
  return name?.slice(0, 2).toUpperCase() || '??'
}

const AVATAR_COLORS = [
  '#1a1a2e', '#16213e', '#0f3460', '#533483',
  '#2b2d42', '#3d405b', '#264653', '#2a9d8f'
]

function avatarColor(name) {
  let hash = 0
  for (const c of name) hash = (hash * 31 + c.charCodeAt(0)) & 0xffffffff
  return AVATAR_COLORS[Math.abs(hash) % AVATAR_COLORS.length]
}

export default function Roasting() {
  const { user } = useAuth()
  const [roastings, setRoastings] = useState([])
  const [comment, setComment] = useState('')
  const [loading, setLoading] = useState(true)
  const [submitting, setSubmitting] = useState(false)
  const [error, setError] = useState('')
  const [submitError, setSubmitError] = useState('')
  const fetchRoastings = async () => {
    try {
      const data = await api.getRoastings()
      setRoastings(Array.isArray(data) ? data.reverse() : [])
    } catch {
      setError('Failed to load roastings')
    } finally {

```

```

    setLoading(false)
  }
}

useEffect(() => {
  fetchRoastings()
  const interval = setInterval(fetchRoastings, 15000)
  return () => clearInterval(interval)
}, [])

const handleSubmit = async (e) => {
  e.preventDefault()
  if (!comment.trim()) return
  setSubmitError('')
  setSubmitting(true)
  try {
    await api.addRoasting(comment)
    setComment('')
    await fetchRoastings()
  } catch (err) {
    setSubmitError(err.message || 'Failed to post')
  } finally {
    setSubmitting(false)
  }
}

return (
  <div className={styles.page}>
    <div className={styles.inner}>
      <div className={styles.header}>
        <h1 className={styles.title}>Roasting Wall</h1>
        <p className={styles.sub}>
          {roastings.length} roast{roastings.length !== 1 ? 's' : ''} and
counting
        </p>
        <p className={styles.sub}>
          Note: admin always visiting this page silently 😊
        </p>
      </div>
      {user ? (
        <form onSubmit={handleSubmit} className={styles.form}>
          <div className={styles.formRow}>
            <div
              className={styles.formAvatar}
              style={{ background: avatarColor(user.username) }}
            >
              {initials(user.username)}
            </div>
            <input type="text" value={comment} />
          </div>
          <button type="submit">Roast</button>
        </form>
      ) : null}
    </div>
  </div>
)

```

```

    </div>
    <div className={styles.formRight}>
      <textarea
        className={styles.textarea}
        placeholder="Drop your roast here..."
        value={comment}
        onChange={e => setComment(e.target.value)}
        maxLength={500}
        rows={3}
      />
      <div className={styles.formFooter}>
        <span className={styles.charCount}>
{comment.length}/500</span>
        {submitError && <span className={styles.submitError}>
{submitError}</span>}
        <button
          type="submit"
          className={styles.postBtn}
          disabled={submitting || !comment.trim()}
        >
          {submitting ? 'Posting...' : 'Post Roast'}
        </button>
      </div>
    </div>
  </form>
) : (
  <div className={styles.loginPrompt}>
    <Link to="/login" className={styles.loginLink}>Sign in</Link> to
drop a roast
  </div>
)}
<div className={styles.feed}>
  {loading ? (
    <div className={styles.empty}>Loading...</div>
  ) : error ? (
    <div className={styles.empty}>{error}</div>
  ) : roastings.length === 0 ? (
    <div className={styles.empty}>No roasts yet. Be the first.</div>
  ) : (
    roastings.map(r => (
      <div key={r.id} className={styles.item}>
        <div
          className={styles.itemAvatar}
          style={{ background: avatarColor(r.roaster) }}
        >

```

```

        {initials(r.roaster)}
      </div>
      <div className={styles.itemBody}>
        <div className={styles.itemMeta}>
          <span className={styles.itemUser}>{r.roaster}</span>
          <span className={styles.itemTime}>{timeAgo(r.created_at)}
        </div>
        <p
          className={styles.itemText}
          dangerouslySetInnerHTML={{ __html: r.comment }}
        />
      </div>
    </div>
  ))
})
</div>
</div>
</div>
)
}

```

- especially in here

```

<p
  className={styles.itemText}
  dangerouslySetInnerHTML={{ __html: r.comment }}
/>

```

- kalau doi pakai ini mah aman cuk harusnya

```

<p>{r.comment}</p>

```

- anyway kenapa bisa ada admin yang dateng ke page yg vuln xss
- itu dateng dari bot, visiting localhost:80 path roasting, 1 menit sekali kalau ga salah

3. Services

HTTP

Frontend

- build frontend vite nya

```
npm run build
[zip file distnya lewat file explorer]
python -m http.server
```

- get filenya dari mesin ubuntu

```
wget [ip]:8000/frontend.zip
```

- download unzip buat unzip code webnya

```
sudo apt install unzip
```

- download apache2 buat deploy frontend

```
sudo apt install apache2
```

- pindahkan file frontend nya

```
unzip frontend.zip
sudo cp -r dist/ /var/www/cave
```

- config apache2 for /var/www/cave

```
sudo nano /etc/apache2/sites-available/cave.conf
```

```
----- di isi
<VirtualHost *:80>
    ServerName 127.0.0.1
    DocumentRoot /var/www/cave

    ProxyPass /api http://127.0.0.1:8080/api
    ProxyPassReverse /api http://127.0.0.1:8080/api

    <Directory /var/www/cave>
        Options -Indexes
        AllowOverride All
        Require all granted
        RewriteEngine On
        RewriteBase /
        RewriteCond %{REQUEST_FILENAME} !-f
        RewriteCond %{REQUEST_FILENAME} !-d
```

```
RewriteRule ^ index.html [L]
</Directory>
</VirtualHost>
```

- restart apache2 and reconf apache2 config

```
sudo a2enmod rewrite proxy proxy_http
sudo a2ensite cave.conf
sudo a2dissite 000-default.conf
sudo systemctl restart apache2
```

- open web frontend nya di device attacker

```
curl http://ip
```

Backend

- build go server nya dengan os linux and arch amd64
- soalnya ini build nya di windows

```
$env:GOOS="linux"; $env:GOARCH="amd64"; go build -o cave_server ./cmd/app/
```

- nyalain python server buat dikirim ke ubuntu

```
python -m http.server
```

- wget file binary server nya, and file db nya

```
wget ip:8000/cave_server
wget ip:8000/data.db
```

- hosting backend biar bisa run terus

```
sudo nano /etc/systemd/system/cave.service
```

- isi config service nya begini

```
[Unit]
Description=Cave Web Service
After=network.target
```



```
[Service]
Type=simple
User=cave_man # ini bisa di isi sesuai run as siapa
WorkingDirectory=/home/cave_man
EnvironmentFile=/home/cave_man/.env
ExecStart=/home/cave_man/cave_server
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

- enable service and start

```
sudo systemctl daemon-reload
sudo systemctl enable cave
sudo systemctl start cave
sudo systemctl status cave
```

Error

- sqlite3 error btw

Binary was compiled with 'CGO_ENABLED=0', go-sqlite3 requires cgo to work

- so i changed my config.go from this

```
package configs

import (
    "database/sql"
    "log"
    _ "github.com/mattn/go-sqlite3"
)

func InitDB() *sql.DB {
    db, err := sql.Open("sqlite3", "./data.db")
    if err != nil {
        log.Fatal(err)
    }
    return db
}
```

- to this

```
package configs

import (
    "database/sql"
    "log"
    _ "modernc.org/sqlite"
)

func InitDB() *sql.DB {
    db, err := sql.Open("sqlite", "./data.db")
    if err != nil {
        log.Fatal(err)
    }
    return db
}
```

Error

- bot chrome error soalnya gada aplikasi chrome di ubuntu server

```
2026/04/21 12:24:30 [bot] admin visiting comment page
2026/04/21 12:24:30 [bot] Error adalah: exec: "google-chrome": executable file
not found in $PATH
```

- jadi download and install dulu aja

```
wget https://dl.google.com/linux/direct/google-chrome-stable_current_amd64.deb
sudo apt install ./google-chrome-stable_current_amd64.deb
```

Error

- ini jwt masih bisa di forge anyway
- kalau gini situasinya xss steal cookie gabisa berlaku
- attacker tinggal gonta ganti octat ke 2 dari jwt token

```
{"id":1,"username":"smithman","isAdmin":false,"exp":1777002454,"iat":1776916054}
```

```
// set ke

{"id":1,"username":"smithman","isAdmin":true,"exp":1777002454,"iat":1776916054}

// then encode base64 and replace

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpbnVlcnQ6bW9uZ2VzZzhvDU

// done jadi admin sekarang gaperlu xss xss an
```

- ternyata masalah ada di auth_middleware.go

```
type contextkey string
const ClaimsKey contextkey = "claims"
type AuthResponse struct {
    Messages string `json:"messages"`
    Token     string `json:"token,omitempty"`
}

func AuthMiddleware(jwtsecret string, next http.HandlerFunc) http.HandlerFunc {
    return func(w http.ResponseWriter, r *http.Request) {
        cookie, err := r.Cookie("kuki")
        if err != nil {
            helper.WriteJSON(w, http.StatusUnauthorized, AuthResponse{
                Messages: "Unauthorized gabole masuk",
            })
            return
        }
        claims := &models.Claims{}
        token, err := jwt.ParseWithClaims(cookie.Value, claims, func(t
*jwt.Token) (any, error) {
            return []byte(jwtsecret), nil
        })
        if err != nil || !token.Valid {
            helper.WriteJSON(w, http.StatusUnauthorized, AuthResponse{
                Messages: "Unauthorized gabole masuk",
            })
            return
        }
        ctx := context.WithValue(r.Context(), ClaimsKey, claims)
        next(w, r.WithContext(ctx))
    }
}
```

```

    }
    [][]byte(jwtsecret), nil
  })

```

- especially in here, change from this

```

token, err := jwt.ParseWithClaims(cookie.Value, claims, func(t *jwt.Token)
(any, error) {
    return [][]byte(jwtsecret), nil
})

```

- to this

```

token, err := jwt.ParseWithClaims(cookie.Value, claims, func(t *jwt.Token)
(any, error) {
    if _, ok := t.Method.(*jwt.SigningMethodHMAC); !ok {
        return nil, fmt.Errorf("unexpected signing method: %v",
t.Header["alg"])
    }
    return [][]byte(jwtsecret), nil
})

```

- lalu di main.go tambahin endpoint baru

```

mux.HandleFunc("GET /api/me", middlewares.AuthMiddleware(jwtSecret, func(w
http.ResponseWriter, r *http.Request) { claims :=
r.Context().Value(middlewares.ClaimsKey).(*models.Claims) helper.WriteJSON(w,
http.StatusOK, claims) }))

```

- lalu di frontend, lib/auth.jsx

```

import { createContext, useContext, useState, useEffect } from 'react'
const AuthCtx = createContext(null)
function parseJWT(token) {
  try {
    const payload = token.split('.')[1]
    return JSON.parse(atob(payload))
  } catch {
    return null
  }
}

export function AuthProvider({ children }) {

```

```

const [user, setUser] = useState(null)
useEffect(() => {
  fetch('/api/me', { credentials: 'include' })
    .then(r => r.ok ? r.json() : null)
    .then(claims => { if (claims) setUser(claims) })
    .catch(() => {})
}, [])
const login = (claims) => setUser(claims)
const logout = () => {
  document.cookie = 'kuki=; Max-Age=0; path=/'
  setUser(null)
}
return (
  <AuthCtx.Provider value={{ user, login, logout }}>
    {children}
  </AuthCtx.Provider>
)
}

export const useAuth = () => useContext(AuthCtx)

```

- login.jsx

```

import { useState } from 'react'
import { Link, useNavigate } from 'react-router-dom'
import { api } from '../lib/api'
import { useAuth } from '../lib/auth'
import styles from './Auth.module.css'
export default function Login() {
  const [form, setForm] = useState({ username: '', password: '' })
  const [error, setError] = useState('')
  const [loading, setLoading] = useState(false)
  const { login } = useAuth()
  const navigate = useNavigate()
  const handleSubmit = async (e) => {
    e.preventDefault()
    setError('')
    setLoading(true)
    try {
      await api.login(form.username, form.password)
      const res = await fetch('/api/me', { credentials: 'include' })
      if (!res.ok) throw new Error('Failed to get user')
      const user = await res.json()
      login(user)
      navigate('/roasting')
    }
  }
}

```



```

        <button type="submit" className={styles.submitBtn} disabled=
{loading}>
            {loading ? 'Signing in...' : 'Sign in →'}
        </button>
    </form>
</div>
</div>
)
}

```

Attack with

- awalan login register dulu
- next cari xss, nanti steal cookie soalnya baca di webnya tulisannya begini

admin web ini rutin bakal datengin page ini lah pokoknya mah

- jadi bisa inject xss steal cookie admin
- daoet cookie admin, ganti cookie nya dah jadi admin
- ada page khusus admin yang vuln lfi
- lfi dah di hardening biar ga vuln sama ../ doang, bypass with//
- next cari /etc/passwd, biar tau usernya siapa aja
- cek2 id_rsa di user2 tersebut, dan cobalah login with id_rsa
- kudu crack juga nanti buat dapetin passphrase

SSH

- setup id_rsa to ssh

generate ssh key with passphrase

- setup nya user yang sekarang yaitu cave_man

```
sudo -u cave_man ssh-keygen -t rsa -b 2048 -f /home/cave_man/.ssh/id_rsa
```

- setup passphrasenya pake pass yang ada di rockyou biar bisa di crack

```
sed -n '69p' /usr/share/wordlists/rockyou.txt
```

[output: teamo]

Enable ssh login with key

- biar bisa langsung connect kalau udah masukin phrase
- tanpa password gitulah ya

```
sudo -u cave_man bash -c 'cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys'
sudo -u cave_man chmod 600 ~/.ssh/authorized_keys
```

Attack with

- crack pass nanti kalau dah dapet id_rsa dari lfi

```
ssh2john id_rsa > id_rsa.hash
john id_rsa.hash --wordlist=/usr/share/wordlists/rockyou.txt
```

- login with id_rsa and passphrase

```
chmod 600 id_rsa
ssh -i id_rsa cave_user@[ip]
```

4. Pivoting

- buat user 2 nya, yang run web ke 2 buat pivoting nantinya

```
sudo useradd -m tribe_leader
```

- set password buat user barusan

```
sudo passwd tribe_leader # set: rahasiainimah
```

- biar user barusan bisa sudo

```
sudo usermod -aG sudo tribe_leader
```

- biar usernya masuk bash, daripada sh kan ya

```
cat /etc/passwd | grep tribe
# cek apakah si tribe masuk shell bash atau sh
```


Turunin Priv user 1

- biar gabisa sudo dia user 1 nya

```
groups cave_man  
sudo deluser cave_man sudo
```

Services HTTP Pivoting

- disini pake cms aja

CVE-2018-7600 (Drupalgeddon2) — RCE unauthenticated

- install deps

```
sudo apt install apache2 php php-mysql php-xml php-gd php-curl php-mbstring  
php-zip mysql-server -y
```

- setup db

```
sudo mysql -u root
```

- made db

```
CREATE DATABASE drupal;  
CREATE USER 'drupal'@'localhost' IDENTIFIED BY 'drupalpass';  
GRANT ALL PRIVILEGES ON drupal.* TO 'drupal'@'localhost';  
FLUSH PRIVILEGES;  
EXIT;
```

- download drupal and setting

```
cd /var/www  
sudo wget https://ftp.drupal.org/files/projects/drupal-7.57.tar.gz  
sudo tar -xzf drupal-7.57.tar.gz
```

- ganti nama, config drupal biar dipegang ama si tribe_leader

```
sudo mv drupal-7.57 internal  
sudo chown -R www-data:www-data /var/www/internal
```

- setup file config apache conf buat service web yang ini

```
sudo nano /etc/apache2/sites-available/internal.conf
```

- setup begini

```
<VirtualHost 127.0.0.1:6060>
    ServerName internal.cave.local
    DocumentRoot /var/www/internal

    <Directory /var/www/internal>
        Options -Indexes
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/internal_error.log
    CustomLog ${APACHE_LOG_DIR}/internal_access.log combined
</VirtualHost>
```

- setup port ke local

```
sudo nano /etc/apache2/ports.conf
```

- tambahin ini

```
Listen 127.0.0.1:6060
```

- begini kira kira

```
root@nolep:/var/www# cat /etc/apache2/ports.conf
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 80
Listen 127.0.0.1:6060

<IfModule ssl_module>
    Listen 443
</IfModule>
```

```
<IfModule mod_gnutls.c>
    Listen 443
</IfModule>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet
```

- enable site nya

```
sudo a2enmod rewrite
sudo a2ensite internal.conf
sudo systemctl restart apache2
```

- buka webnya pake ssh local port forwarding
- di mesin lain, misal kali linux attacker

```
ssh -L 6060:127.0.0.1:6060 cave_man@[ip_ubuntu_server]
```

- di browser attacker buka aja 127.0.0.1:6060

Setup CMS di browser

- nanti setup aja begini

```
- Choose profile: pilih standard
- Choose language: english built in
```

- di bagian verify requirements nya belum centang semua, err disini

```
|Web server|Apache/2.4.29 (Ubuntu)| |
|OK|PHP|7.2.24-0ubuntu0.18.04.17|
|OK|PHP register globals|Disabled|
|OK|PHP extensions|Enabled|
|OK|Database support|Enabled|
|OK|PHP memory limit|128M|
```

```
|Error|File system||
```

|The directory _sites/default/files_ does not exist. An automated attempt to create this directory failed, possibly due to a permissions problem. To proceed with the installation, either create the directory and modify its permissions manually or ensure that the installer has the permissions to

create it automatically. For more information, see INSTALL.txt or the [online handbook](http://drupal.org/server-permissions).| | |

|OK|Unicode library|PHP Mbstring Extension|

|Error|Settings file|The settings file does not exist.|

|The Drupal installer requires that you create a settings file as part of the installation process. Copy the `../sites/default/default.settings.php` file to `../sites/default/settings.php`. More details about installing Drupal are available in [INSTALL.txt](http://127.0.0.1:6060/INSTALL.txt).| | |

- solve with :

```
sudo mkdir -p /var/www/internal/sites/default/files
sudo chown www-data:www-data /var/www/internal/sites/default/files
sudo chmod 755 /var/www/internal/sites/default/files
```

- dan ini :

```
sudo cp /var/www/internal/sites/default/default.settings.php
/var/www/internal/sites/default/settings.php

sudo chown www-data:www-data /var/www/internal/sites/default/settings.php
sudo chmod 666 /var/www/internal/sites/default/settings.php
```

- next lanjut config

- Set up database:

Database name: drupal
Database username: drupal
Database password: drupalpass
Host: localhost
Port: 3306

- Install profile: ditunggu aja

- Configure site:

Site name: Internal Cave Portal
Site e-mail: admin@cave.local
Username: admin
E-mail: admin@cave.local
Password: admin2026!
Confirm password: admin2026!

- done, cek di browser 127.0.0.1:6060

Test exploit

- udah ssh -L ke target, and check webnya
- lanjut exploit pakai msf

```
msfconsole
search CVE-2018-7600
set set aja
exploit
```

- nah ternyata ada error, as www-data kita bisa masuk folder home user2 disitu

```
ls -la /home
drwxr-xr-x  8 cave_man      cave_man      4096 May  3 11:50 cave_man
drwxr-xr-x  2 tribe_leader tribe_leader    4096 Apr 24 14:55 tribe_leader
```

- nah dia begini makanya bisa di read

```
drwxr-xr-x
```

- set permission aja

```
chmod 700 /home/cave_man
chmod 700 /home/tribe_leader

drwx-----  8 cave_man      cave_man      4096 May  3 11:50 cave_man
drwx-----  2 tribe_leader tribe_leader    4096 Apr 24 14:55 tribe_leader
```

Setup permission

- kan doi jadi www-data ya after exploit tuh
- kira2 gimana ya setup privesc www-data ke tribe_leader
- baca config di ini aja nanti attackernya

```
cat /var/www/internal/sites/default/settings.php | grep password
```

- sengaja kasih comment ke creds disitu buat password asli nya

- tapi root gabisa edit permission nya gini

```
root@nolep:/home/cave_man ls -la /var/www/internal/sites/default/settings.php
-r--r--r-- 1 www-data www-data 26559 May  2 16:12
/var/www/internal/sites/default/settings.php
```

- yauda edit dulu

```
chmod 644 /var/www/internal/sites/default/settings.php
vim /var/www/internal/sites/default/settings.php
```

- kasih beginian

```
$databases = array (
  'default' =>
  array (
    'default' =>
    array (
      'database' => 'drupal',
      'username' => 'drupal',
      # 'password' => 'rahasiainimah',
      'password' => 'drupalpass',
      'host' => 'localhost',
      'port' => '3306',
      'driver' => 'mysql',
      'prefix' => '',
    ),
  ),
);
```

- balikin biar readonly

```
chmod 444 /var/www/internal/sites/default/settings.php
```

```
root@nolep:/home/cave_man ls -la /var/www/internal/sites/default/settings.php
-r--r--r-- 1 www-data www-data 26598 May  3 12:19
/var/www/internal/sites/default/settings.php
```

- isolasi path /var/www/internal biar gabisa di akses cave_man

```
sudo chmod 750 /var/www/internal
```

Privilege Escalation

SUID binary

- setup dulu begini
- sudo su ke root dulu as root aja run nya

```
cp /bin/bash /usr/local/bin/bash_suid
chown root:root /usr/local/bin/bash_suid
chmod u+s /usr/local/bin/bash_suid

ls -la /usr/local/bin/bash_suid
```

- error, bisa di run as cave_man jir, harusnya bisa as tribe_leader aja
- gini in aja

```
chown root:tribe_leader /usr/local/bin/bash_suid
chmod 4750 /usr/local/bin/bash_suid

ls -la /usr/local/bin/bash_suid
```

- nanti attacker enum gini as tribe_leader

```
find / -perm -u=s -type f 2>/dev/null
```

- nice tinggal turuin tribe_leader biar gabisa sudo su

```
sudo deluser tribe_leader sudo
```

5. Specs and Detail VM

Name :

- ubuntu-18.04.6-live-server-amd64

Spek :

- ram: 1024mb

- cpu: 1core
- memory: 10,59 GB

Flags :

- user.txt: FLAG{a323e8117e4fffd8f9c7db240eebf9ea}
- root.txt: FLAG{6147e882471ffb950f0520bc028ca2c8}

walkthrough - video

- <https://youtu.be/YJdd1gI92lk>